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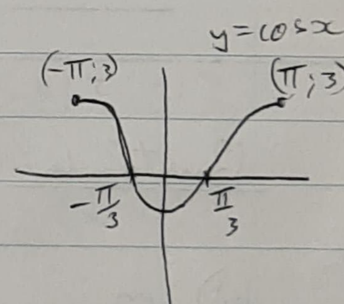
Supplementary Exam

Question 1:

1.1) $x \in [3; 5)$
1.2) 5
1.3) 14
1.4) Horizontal asymptote at $x = 0$
1.5) 20
1.6) $\arctan\left(\frac{x}{2}\right) + c$
1.7) $x \in [3; 5]$
1.8) $a = e^2$
1.9) $x = \frac{1}{2}$
1.10) $y = x - 3$

Question 2:



2.1) for f increasing: $f'(x) > 0$

$f'(x) = 1 - 2\cos x$

$0 < 1 - 2\cos x$

$1 < 2\cos x$

$\frac{1}{2} < \cos x$

$\cancel{K\pi}\ \pi K \pm \frac{\pi}{3} < x$ → Because $\cos x$ is positive in quadrants 1 & 4

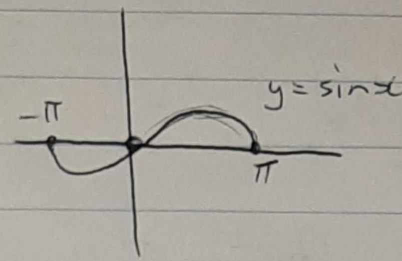
$\therefore x < -\frac{\pi}{3}$

$x > \frac{\pi}{3}$ but $x \in (-\pi; \pi)$

$\Rightarrow x \in \left(-\pi; -\frac{\pi}{3}\right) \cap \left(\frac{\pi}{3}; \pi\right)$

2.2) for concave upwards: $f''(x) > 0$

$$f''(x) = \sin x$$
$$0 < \sin x$$
$$0 < x$$
$$\Rightarrow x \in (0; \pi)$$



Question 3:

$$x = 2\ln(\cancel{*}\, y + a)$$
$$\cancel{e^{x} = 2(y+a)}$$
$$e^{x} = 2\left(e^{\ln(y+a)}\right)$$

$$e^{x} = 2(y+a)$$
$$\frac{e^{x}}{2} - a = y$$

$$\Rightarrow f^{-1}(x) = \frac{e^{x}}{2} - a$$

Question 4:

4.1) $m_{tangent} = \frac{dy}{dx}$ in the point $(x; y)$

$$xy + y^3 = \sec x$$
$$y + x.y' + 3y^2.y' = \sec x . \tan x$$
$$y'(x + 3y^2) = \sec x . \tan x - y$$
$$y' = \frac{\sec x . \tan x - y}{x + 3y^2}$$
$$\therefore \frac{dy}{dx} = \frac{\sec x . \tan x - y}{x + 3y^2}$$
$$\Rightarrow m_{tangent line} = \frac{\sec x . \tan x - y}{x + 3y^2}$$

4.2) $m_{tangentline} = \frac{dy}{dx}$ in the point $x = 1$

$$y = (x^2 + x)^x$$

$\ln y = x \ln(x^2 + x)$ (logarithmic differentiation)

$$\frac{1}{y} \cdot y' = \ln(x^2 + x) + x\left(\frac{2x+1}{x^2+x}\right)$$

$$= \ln(x^2+x) + \frac{2x^2+x}{x^2+x}$$

$$= \ln(x^2+x) + \frac{2x+1}{x+1}$$

but $y' = (x^2+x)^x \left(\ln(x^2+x) + \frac{2x+1}{x+1}\right)$

$\Rightarrow m_{tangent} = 2\left(\ln(2) + \frac{3}{2}\right)$

$$= 2\ln(2) + 3$$

Question 5:

$\bar{OA} = \langle 1; -1; 1 \rangle$

$\bar{OB} = \langle 3; 2; -1 \rangle$

$$\frac{\langle 1; -1; 1 \rangle \times \langle 3; 2; -1 \rangle}{|\bar{OA}|\,|\bar{OB}|}$$

$$\begin{vmatrix} i & j & k \\ 1 & -1 & 1 \\ 3 & 2 & -1 \end{vmatrix}$$ (crossproduct of 2 vectors)

$$= \frac{\langle -1; 4; 5 \rangle}{\sqrt{42}}$$

$$= \left\langle -\frac{1}{\sqrt{42}}; \frac{4}{\sqrt{42}}; \frac{5}{\sqrt{42}} \right\rangle$$

$= i(-1) - j(-4) + k(5)$

$= \langle -1; 4; 5 \rangle$

Test: $\left\langle -\frac{1}{\sqrt{42}}; \frac{4}{\sqrt{42}}; \frac{5}{\sqrt{42}} \right\rangle \cdot \langle 1; -1; 1 \rangle$

$$= -\frac{1}{\sqrt{42}} + -\frac{4}{\sqrt{42}} + \frac{5}{\sqrt{42}}$$

$$= 0$$

$\Rightarrow \left\langle -\frac{1}{\sqrt{42}}; \frac{4}{\sqrt{42}}; \frac{5}{\sqrt{42}} \right\rangle$ is indeed a orthogonal vector to both $\bar{OB}$ and $\bar{OA}$

Question 6:

6.1) $\lim_{x \to \infty^-} x^2 e^x \quad (\infty \cdot \infty)$

$= \lim_{x \to \infty^-} 2x \cdot e^x$ (L'Hospital's rule)

$= \lim_{x \to \infty^-} 2 \cdot e^x$ (L'Hospital's rule)

$=$

6.1) let $\lim_{x\to\infty} y = \lim_{x\to\infty} x^2 e^x$

$\lim_{x\to\infty^-} \ln y = \lim_{x\to\infty^-} 2\ln x + \lim_{x\to\infty^-} x$

$= \lim_{x\to\infty^-} \frac{2}{x} + 1$ (L'Hospital's rule)

$= 0 + 1$

$= 1$

$\Rightarrow y = e^1 = e$

6.2) let $y = \lim_{x\to 0^+} x^x$

$\lim_{x\to 0^+} y = \lim_{x\to 0^+} x^x$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} x \ln x$

$= \lim_{x\to 0^+} \ln x + \frac{x}{x}$ (L'Hospital's rule)

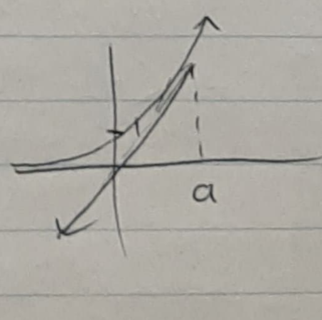
$= \lim_{x\to 0^+} \ln x$

$y = \lim_{x\to 0^+} e^x$

$= 1$

Question 7:

$(0;0)$



$m_{tang} = f'(x) = e^x$

$= e^a$

$y = mx$

$=$